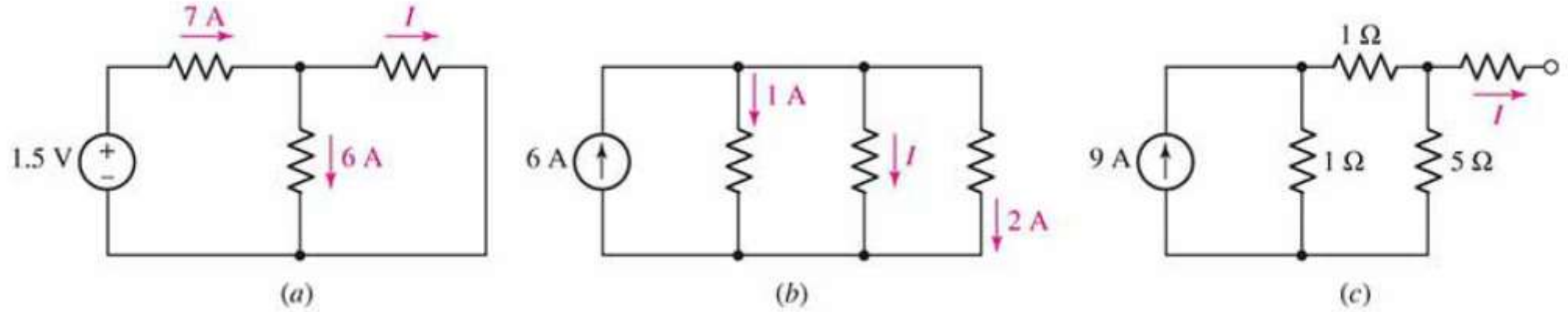
1. Use Kirchhoff's Current Law and Kirchhoff's Voltage Law to solve the following questions.

(1). Determine the current labeled I in each of the circuits below.[15]



(a) 7-6-I =0, I =1 A

(b) 6-1-I-2=0, I=3 A

(c) I =0

63. Consider the seven-element circuit depicted in Fig. 3.99. (*a*) How many nodes, loops, and branches does it contain? (*b*) Calculate the current flowing through each resistor. (*c*) Determine the voltage across the current source, assuming the top terminal is the positive reference terminal.

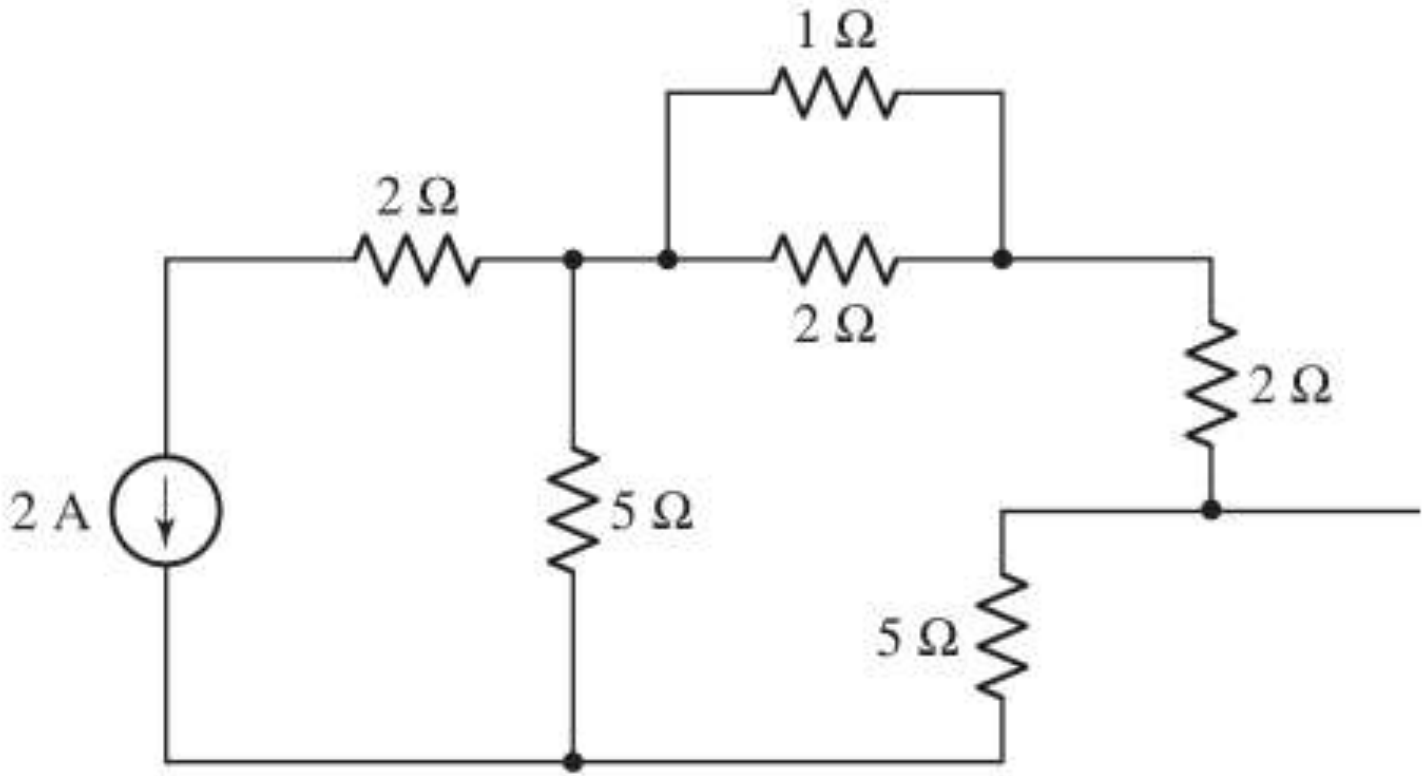


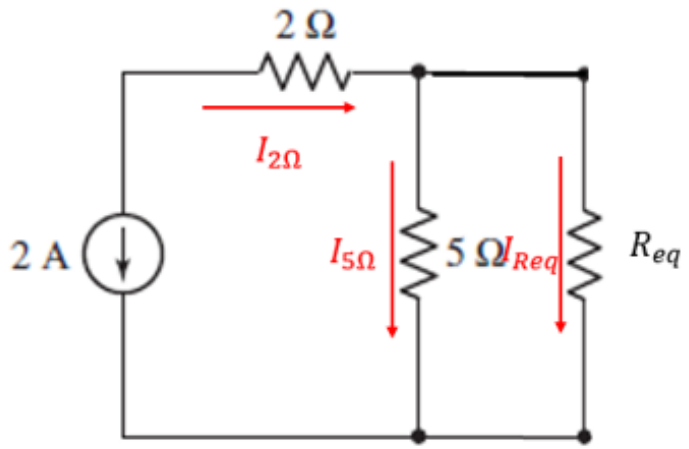
FIGURE 3.99

Answer:

(a) 5 nodes; 6 loops; 7 branches;

(b) Define all currents as flowing either left to right or downward.

Frist, combine the resistors

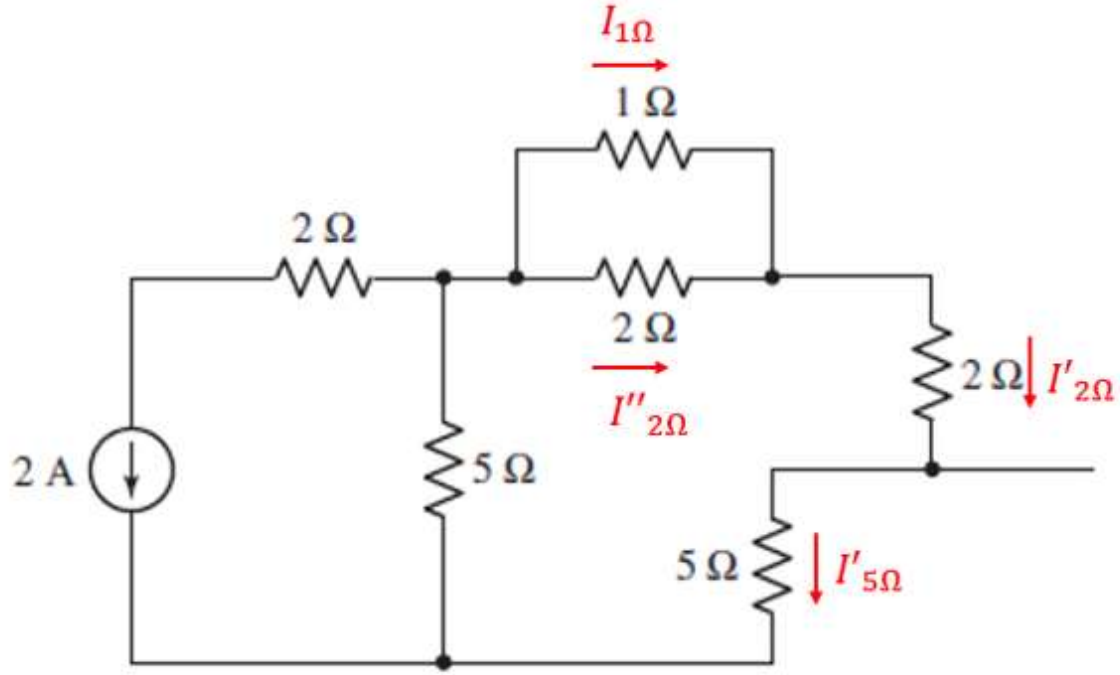


Note the combination $R_{eq} = 1//2 + 2 + 5 = 7.667\ \Omega$.

By inspection, $I_{2\Omega}$ = –2 A;

Apply current division:

$I_{5\Omega}$ = (-2)(7.667)/(5 + 7.667) = –1.21 A; I_{Req} = (-2)(5)/(5 + 7.667) = –790 mA;



$I'_{5\Omega} = I'_{2\Omega} = I_{Req} = -790$ mA;

The remaining currents are found by current division:

$I_{1\Omega} = -0.7895(2)/3 = -526$ mA;

$I''_{2\Omega} = -0.7895(1)/3 = -263$ mA

(c) $V_{2A} = V_{2\Omega(left)} + V_{5\Omega(left)} = (-2)(2) + (-1.211)(5) = -10.06$ V.

24. The circuit shown in Fig. 3.65 includes a device known as an op amp. This device has two unusual properties in the circuit shown: (1) $V_d = 0$ V, and (2) no current can flow into either input terminal (marked "−" and "+" inside the symbol), but it *can* flow through the output terminal (marked "OUT"). This seemingly impossible situation—in direct conflict with KCL—is a result of power leads to the device that are not included in the symbol. Based on this information, calculate V_{out}. (*Hint*: Two KVL equations are required, both involving the 5 V source.)

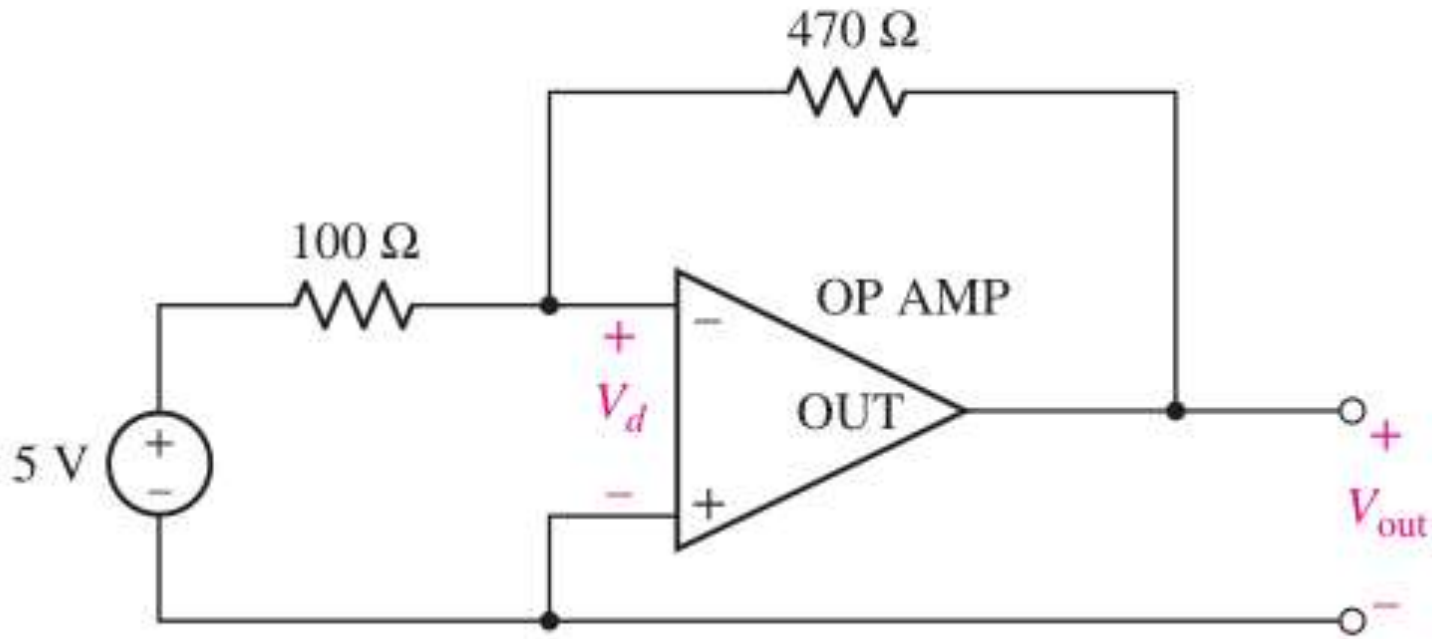


FIGURE 3.65

Answer:

Define the current I flowing out of the "+" reference terminal of the 5 V source. This current also flows through the 100 Ω and 470 Ω resistors since no current can flow into the input terminals of the op amp.

Then, $-5 + 100I + 470I + v_{out} = 0$ [1]

Further, since $V_d = 0$, we may also write

$-5 + 100I = 0$ [2]

Solving, $V_{out} = 5 - 570I = 5 - 570(5/100) = \boxed{-23.5\ V}$